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INTERNSHIP TASK REPORT

TASK 03

OUT-OF-SPECIFICATION (OOS) INVESTIGATION REPORT

Evidence-based analysis of a published pharmaceutical blend-uniformity investigation

Document control	Details
Internship	UpStartInterns – Pharmaceutical Quality Control
Task	Task 03 – Case study: Out-of-Specification result
Primary case source	Jain SK, Jain RK. Investigations and CAPA: Quality system for continual improvement in pharmaceutical industry. International Journal of Research in Pharmacy and Pharmaceutical Sciences. 2017;2(6):47–54.
Case subject	Anonymised XYZ Tablets USP; blend-uniformity (BU) OOS investigation
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Document status	Training case analysis using published numerical data; not an original GMP record
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DATA-INTEGRITY STATEMENT

Numerical results and investigation findings in this report are transcribed or summarised from the cited publication. Product and batch identifiers are anonymised in the source. No experimental readings, batch records or manufacturing parameters have been invented. Interpretations beyond the authors' reported conclusions are identified as analysis.

1. Executive summary

This report analyses a published pharmaceutical blend-uniformity (BU) investigation involving six batches of anonymised XYZ Tablets USP. The article reports low BU results and describes a systematic investigation involving laboratory checks, manufacturing-record review, hypothesis testing, HPLC measurements and CAPA.

The investigation identified two laboratory preparation concerns. First, some initially diluted sample solutions had not been adequately mixed, and remixing produced substantially higher results. More importantly, certain newly introduced 10 mL volumetric flasks (designated "Glassware A") caused loss of API when water was used as diluent. Experiments demonstrated that the API was adsorbed onto the inner surface of the affected flasks. The effect was not observed when phosphate buffer/acetonitrile (85:15, v/v) was used.

The manufacturing batch records were reviewed for raw-material quantities, API potency calculations, manufacturing process, critical parameters and yields; the authors reported these checks as satisfactory and did not identify a manufacturing root cause. The final assignable cause was laboratory-related adsorption of API by specific glassware under the BU method's water-diluent condition. The paper reports that subsequent BU results were within specification following corrective actions.

Key question	Finding
Was there a genuine low BU/OOS signal?	Yes. The publication reports low BU results across the investigated batches.
Was a manufacturing cause identified?	No. The reviewed manufacturing records and critical process checks were reported satisfactory.
Was a laboratory cause demonstrated?	Yes. Adsorption of API onto specific 10 mL volumetric flasks in the presence of water was demonstrated experimentally.
Was the cause supported by data?	Yes. Comparative dilution, holding-time and API-recovery experiments supported adsorption.
Final investigation conclusion	The low BU results were attributed to a laboratory artefact caused by API adsorption in specific glassware when water was used as diluent.
CAPA effectiveness	The authors report compliant results in the next 10 batches and review of the next 50 batches with results within specification.

2. Case background and OOS definition

Blend uniformity evaluates the distribution of active pharmaceutical ingredient (API) within a powder blend before downstream dosage-form manufacture. A low result from a blend sample may indicate actual non-uniformity or an analytical/sample-preparation artefact. The investigation must distinguish these possibilities using documented evidence.

The source article describes BU results for six batches of anonymised XYZ Tablets USP. The stated acceptance criteria were: mean of all test results between 95.0% and 105.0%, and relative standard deviation (RSD) not more than 5.0%. The source does not disclose the product's active ingredient, strength, actual commercial batch identifiers or full manufacturing records.

Batch	Mean (%)	Min (%)	Max (%)	Source interpretation
1	83.3	49.3	101.0	Reported among the non-compliant BU batch results
2	95.9	73.6	101.7	Included in the investigated dataset
3	91.4	60.8	99.9	Reported among the non-compliant BU batch results
4	97.9	77.1	101.7	Included in the investigated dataset
5	99.1	83.2	103.1	Included in the investigated dataset
6	97.5	83.2	101.0	Included in the investigated dataset
Specification	Mean 95.0 – 105.0	—	—	RSD: NMT 5.0%

Interpretation caution : The paper presents the six-batch table as BU results that did not comply, but the displayed table does not provide the batch-wise RSD values. Therefore, this report does not independently reconstruct each batch's complete pass/fail status from mean/minimum/maximum alone. The publication's narrative is retained as the source for the investigation scope.

3. Investigation objectives and scope

- To determine whether low BU results were caused by actual blend non-uniformity or laboratory/sample-preparation error.
- Review of analytical execution, sample preparation, glassware and diluent effects using the reported experiments.
- Review of manufacturing records and historical BU/CU performance for evidence of process-related causes.
- Identifying the assignable root cause and assess the evidence supporting it.

- To summarise corrective actions and the reported effectiveness checks.

4. Phase I – Laboratory investigation

4.1 Initial laboratory assessment

The authors report that the initial laboratory investigation found no analytical error, human error, instrument error or material error. The team then investigated sample-preparation hypotheses because low BU results continued to occur.

Laboratory area	Investigation described in publication	Finding
Analytical / human error	Initial laboratory assessment	No analytical or human error identified in the initial assessment.
Instrument error	Initial laboratory assessment	No instrument error identified.
Material error	Initial laboratory assessment	No material error identified.
Sample-solution mixing	Remixing selected stock/sample solutions and retesting	Results increased markedly, showing that some diluted solutions had not been adequately mixed.
Glassware compatibility	Compare results in different 10 mL volumetric flasks and diluents	Specific new glassware caused low results with water as diluent.
Adsorption / recovery	Hold standard solution in suspected flasks, then recover API from flask surface	API was recovered from the affected flasks, supporting adsorption.

4.2 Hypothesis 1 – Incomplete mixing of diluted samples

The team hypothesised that insufficient mixing of diluted sample solutions could cause low BU readings. Selected solutions were remixed and reanalysed.

Sample	Initial result (%)	After mixing (%)	Change (percentage points)
Batch 1 / S1	89.3	98.8	+9.5
Batch 1 / S3	86.9	98.6	+11.7
Batch 1 / S6	46.7	100.2	+53.5

These results support incomplete mixing as a source of low results in the selected preparations. In a further experiment, remixed samples from Batch 1 gave S1: 88.8% to 99.4%; S2: 99.9% to 101.0%; S3: 92.3% to 99.2%; and S6: 49.3% to 100.7%. However, the investigation continued because repeated low values still required explanation.

4.3 Hypothesis 2 – Adsorption or degradation in specific glassware

The authors report that results from some samples fell further after transfer into fresh vials. Inspection identified newly introduced volumetric flasks with a porous inner surface, designated “Glassware A”.
A homogeneous sample solution gave 101.0% when tested from a 50 mL flask. After transfer into three suspected 10 mL flasks and one-hour holding, the results were 72.3%, 66.0% and 69.9%.

This comparison suggested that the low result was associated with the specific 10 mL glassware rather than the original blend sample alone.

4.4 Hypothesis 3 – Influence of diluent

The BU method used water as diluent, whereas the content-uniformity (CU)/assay method used pH 7.4 phosphate buffer and acetonitrile in an 85:15 (v/v) ratio. The team tested both glassware types with the buffer/acetonitrile diluent. The published results remained around 102–106% over four hours, and the authors reported no degradation in either glassware type under this diluent condition.

Flask group	Flask 1 (0/2/4 h)	Flask 2 (0/2/4 h)	Flask 3 (0/2/4 h)
Previously low-result glassware A	103.3 / 103.9 / 102.5	103.4 / 103.5 / 102.3	104.6 / 106.4 / 105.0
Previously passing glassware B	103.4 / 104.3 / 103.1	103.4 / 103.4 / 102.2	103.4 / 103.6 / 102.3

The result pattern supported a diluent-dependent effect: the suspect glassware produced low values when water was used, but not when phosphate buffer/acetonitrile was used.

4.5 Hypothesis 4 – Direct API adsorption and recovery experiment

The team investigated whether API was retained on the inner surface of the suspected flasks. Standard solution was held in two low-result flasks and one passing flask. After four hours, the solution was decanted; water was added and the flasks were heated at 60°C for 30 minutes to recover retained API.

Flask	Initial API (mg/10 mL)	After 4 h (mg/10 mL)	API recovered from flask	Total API (A+B)	Reported total
1 – low-result flask A	0.129	0.120 (92.3%)	0.0050 mg/5 mL (3.9%)	0.1250 mg	96.2%
2 – low-result flask A	0.127	0.111 (85.4%)	0.0095 mg/5 mL (7.3%)	0.1205 mg	92.7%
3 – passing flask B	0.131	0.131 (100.8%)	0.000 mg (0%)	0.131 mg	100.8%

Recovery of API from the two suspect flasks, together with no recovered API from the passing flask, supported adsorption onto the inner surface of the specific glassware. This was the key confirmatory experiment reported by the authors.

4.6 Phase I conclusion

Assignable laboratory cause established: API adsorption onto specific 10 mL volumetric flasks when water was used as the BU-test diluent. Inadequate mixing of selected diluted samples was also demonstrated as an additional sample-preparation issue.

The conclusion is supported by comparative glassware testing, diluent comparison, time-dependent results and recovery of API from the suspect flask surfaces. The investigation did not rely solely on a passing repeat result.

5. Phase II – Manufacturing investigation

The publication reports that the executed batch manufacturing records for all investigated batches were reviewed. The review covered raw-material quantities dispensed, API potency calculations, the manufacturing process and parameters, and batch yields at critical stages. These were reported as satisfactory. The authors therefore ruled out the probability of a manufacturing error involving process, personnel or machinery in the investigated case.

Manufacturing review area	Evidence reviewed (as reported)	Reported outcome
Raw-material dispensing	Quantity of raw material dispensed	Satisfactory
API potency calculation	Potency calculation used for dispensing / batch calculation	Satisfactory
Manufacturing process	Process and manufacturing steps	Satisfactory

Critical process parameters	Parameters at critical manufacturing stages	Satisfactory
Yield	Batch yield at critical stages	Satisfactory
Historical quality trend	Previous BU and CU results	One similar low BU event had previously been attributed to dilution error; no CU OOS was reported.

Important limitation: the article summarises the manufacturing-record review but does not reproduce actual batch manufacturing record pages, individual process parameter values, equipment logbooks or raw-material lot identifiers. Accordingly, this report can state what the authors reported reviewing and their conclusion, but cannot independently audit those confidential records.

5.1 Phase II conclusion

No manufacturing root cause was identified in the published case. The authors reported satisfactory review of dispensing, API potency, process parameters and critical-stage yields. The experimentally demonstrated cause was located in laboratory sample preparation and glassware compatibility, not in the manufacturing process.

6. Root-cause analysis

Potential cause	Evidence	Assessment
Actual blend non-uniformity	Manufacturing records were reported satisfactory; low values tracked with sample preparation and specific glassware.	Not supported as the root cause in the published investigation.
Incomplete mixing of diluted sample	Remixing selected solutions raised results substantially, e.g. 46.7% to 100.2%.	Confirmed contributing laboratory preparation error for selected solutions.
Instrument / analyst / material error	Initial laboratory assessment reported no instrument, human, analytical or material error.	Not identified in initial assessment.
API adsorption onto glassware A	Low values in specific 10 mL flasks with water; API recovered from affected flask surfaces.	Confirmed principal root cause reported by the authors.
Manufacturing process error	Batch records, raw material quantities, potency calculations, process parameters and yields reported satisfactory.	Not identified.

Final root-cause statement : The low BU results were caused by adsorption of API onto the inner surface of certain newly introduced 10 mL volumetric flasks when water was used as the diluent. Incomplete mixing of selected diluted sample solutions also produced artificially low results during the investigation. The published evidence did not identify a manufacturing-process cause.

7. CAPA and effectiveness verification

Action	Type	Evidence / effectiveness reported
Discontinue the manufacturer/source of Glassware A that produced low results.	Corrective	The source article reports discontinuation of the glassware manufacturer.
Change the BU-test diluent from water to pH 7.4 phosphate buffer/acetonitrile (85:15, v/v), aligning it with the assay/CU diluent.	Corrective	Comparative experiment showed no low-result effect in either glassware group with this diluent.
Use newly received, segregated glassware for BU analysis while implementing the approved method.	Correction / control	The authors report the next 10 batches gave BU results within specification.
Review subsequent BU results to verify sustained effectiveness.	Effectiveness check	The next 50 batches were reviewed; the article reports all results within specification.

Published results for the next 10 batches after corrective actions:

Batch	Average (%)	Minimum (%)	Maximum (%)	RSD (%)
1	102.5	101.4	106.6	1.5
2	102.3	101.3	103.5	0.8
3	100.3	98.6	101.4	0.9
4	102.0	99.9	105.1	1.8
5	102.6	101.3	106.3	1.6
6	102.8	100.7	104.6	1.1
7	103.1	100.9	104.8	1.1

8	102.3	100.2	104.1	1.0
9	103.2	100.5	104.8	1.2
10	103.6	101.6	105.6	1.4

8. Impact assessment

- **Laboratory result integrity :** BU results generated using the implicated glassware with water as diluent may be biased low and require documented retrospective assessment.
- **Batch quality :** The evidence supports an analytical artefact for the investigated low results; the paper reports satisfactory manufacturing-record review. Actual disposition of each original batch remains subject to the firm's authorised Quality Unit and complete records.
- **Other batches :** Historical and subsequent BU results should be reviewed for use of the same glassware source, diluent, analyst practices and method version.
- **Other methods :** Assess whether any other laboratory test uses the same glassware/diluent combination and could be affected by adsorption.
- **CAPA effectiveness :** The reported next-10-batch data and next-50-batch review support effectiveness, as reported by the authors.

9. Final conclusion and OOS disposition

The published investigation provides a scientifically supported laboratory assignable cause for the low BU results: adsorption of API onto specific 10 mL volumetric flasks when water was used as the diluent. The finding was supported by direct comparative experiments and recovery of API from the flask surface. A separate sample mixing issue was also demonstrated for selected diluted solutions.

The Phase II manufacturing review, as reported by the authors, did not identify an error in raw-material dispensing, API potency calculations, process parameters or critical-stage yields. The evidence

therefore points to laboratory sample-preparation/glassware effects rather than a manufacturing-process failure.

Disposition conclusion for this case study : The initial low BU results are attributable to a demonstrated laboratory artefact and may be scientifically invalidated only for the affected analyses/batches after the firm's Quality Unit documents the exact affected scope, traceability to the implicated glassware and method conditions, and approves the investigation under its SOP. The paper reports corrective actions and subsequent compliant results; it does not provide the original controlled batch-disposition records.

This distinction is important: a training report can conclude that the published evidence supports laboratory invalidation, but it must not claim to know the exact batch-release decisions unless the source explicitly reports them.

10. Investigation decision flow

1. Recording the original BU OOS result and preserve all original analytical data and sample preparations.
2. **Conducting Phase I laboratory investigation :** analyst, calculations, instrument, standards, reagents, glassware, sample preparation, dilution and method execution.
3. **Testing hypotheses using controlled comparisons**. In this case, remixing selected samples improved results; suspect glassware and diluent comparisons showed a further effect.
4. **Confirmation of root cause using direct evidence**. In this case, API was recovered from suspect glassware after contact with the solution.
5. **Conducting Phase II manufacturing review when required :** dispensing, potency correction, process parameters, yields, equipment, deviations and historical trends.
6. Documenting the evidence-based root cause, affected scope, impact assessment, CAPA and effectiveness checks.

7. Quality Unit decides whether the original OOS can be invalidated for the affected tests and makes the final batch disposition.

11. Limitations and data-integrity notes

- The source is an anonymised published case study; product identity, API identity, strength and actual batch identifiers are not disclosed.
- The article does not provide all original chromatograms, raw instrument files, full batch manufacturing records or exact process parameter values.
- The initial six-batch table provides mean, minimum and maximum but not batch-wise RSD values. This report does not independently recalculate all initial batch acceptance decisions.
- The source reports that the next 50 batches were within specification, but the detailed numerical table shown in the article covers the next 10 batches.
- No missing product or batch data have been invented in this report.
- This is an educational retrospective analysis, not a GMP-controlled OOS record or a claim that the author performed the experiments.

12. References

- Jain SK, Jain RK. Investigations and CAPA: Quality system for continual improvement in pharmaceutical industry. International Journal of Research in Pharmacy and Pharmaceutical Sciences. 2017;2(6):47–54, especially pp. 50–53. Full-text PDF: https://www.pharmacyjournal.in/assets/archives/2017/vol2issue6/2-6-21-513.pdf
- U.S. Food and Drug Administration. Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production: Guidance for Industry. May 2022. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/investigating-out-specification-oos-test-results-pharmaceutical-production-level-2-revision

- U.S. Food and Drug Administration. Risk-Based Blend and Content Uniformity Assessment: A Case Study. FDA Science Forum 2023. https://www.fda.gov/media/168894/download
- U.S. Code of Federal Regulations. 21 CFR 211.192 – Production record review. https://www.ecfr.gov/current/title-21/section-211.192

13. README – source selection and report decisions

Decision	Explanation
Why this case was selected	It is an openly accessible published case with actual numerical results, laboratory hypothesis testing, a manufacturing-record review, a demonstrated root cause, CAPA and follow-up effectiveness data.
Primary data source	Jain & Jain (2017), pages 50–53.
Why not use Lupin as the main case	The FDA warning letter is valuable but redacts numerical details and does not provide a completed manufacturing investigation.
Use of FDA sources	FDA guidance is used as the regulatory framework; the FDA 2023 poster is a supplementary example of risk-based BU/CU assessment, not the source of this case's root cause.
Data treatment	Numerical values are reproduced from the article; anonymised identifiers are retained. No simulated numerical dataset is included.
Final conclusion	Laboratory artefact demonstrated; manufacturing root cause not identified; affected analytical results may be invalidated only with documented scope and Quality Unit approval.

— END OF REPORT —